SOC Brute Force Detection Report

🔹 Introduction

This project demonstrates how a brute force attack can be detected using Windows Security Event Logs.

The analysis focuses on identifying failed and successful login attempts using Event IDs 4625 and 4624.

🔹 Lab Setup

The lab environment consists of:

- Windows Server 2019 (Active Directory)
- Windows 11 Client machine
- A user account named "Abdullah"
- VMware virtual environment

This setup simulates a basic enterprise network.

🔹 Attack Scenario

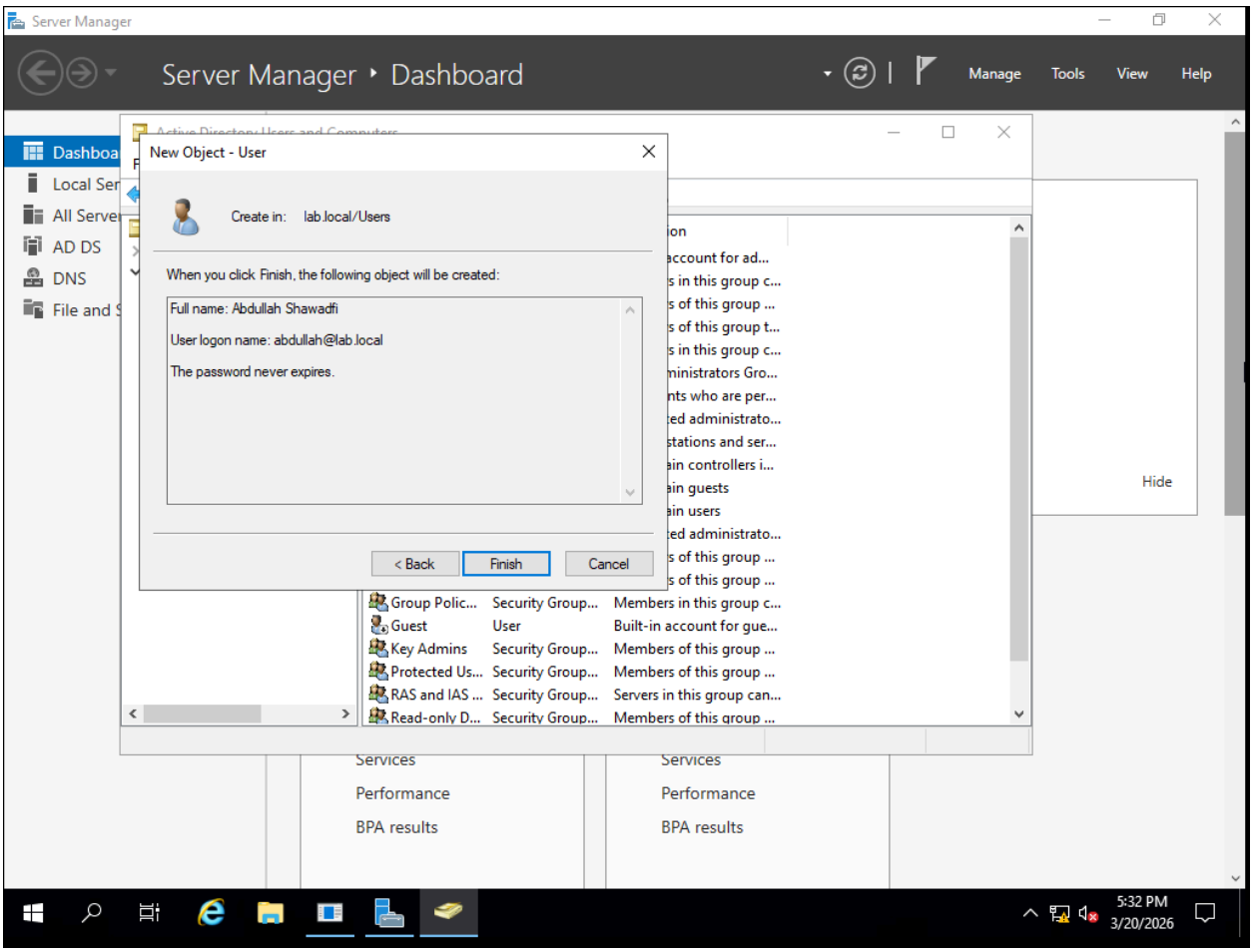
During log analysis, multiple failed login attempts (Event ID 4625) were observed targeting the user "Abdullah".

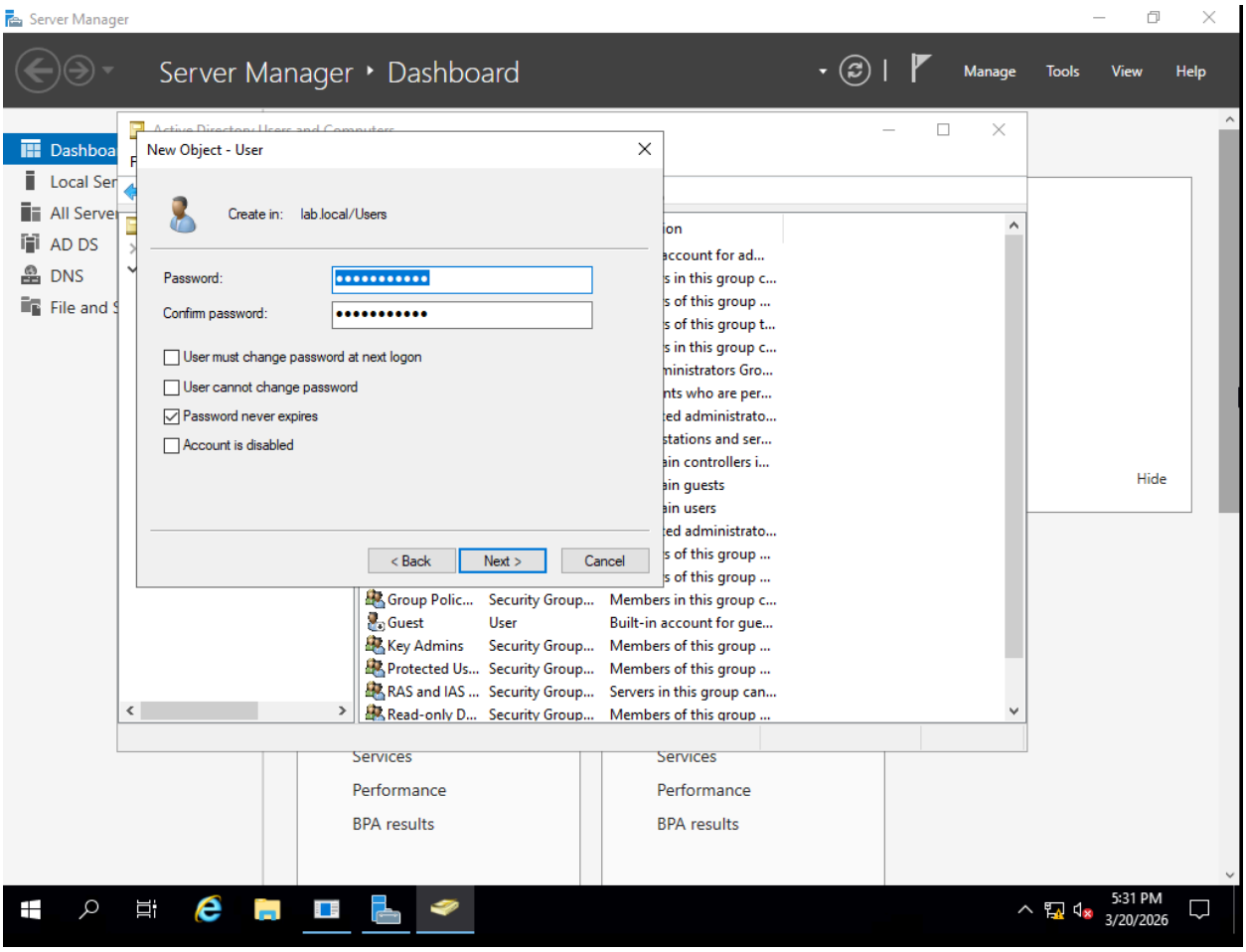
These attempts occurred within a short time period, which indicates a possible brute force attack.

Shortly after these failed attempts, a successful login (Event ID 4624) was recorded for the same user account.

🔹 Evidence

These image shows the user account "Abdullah" created in Active Directory, which was used to simulate login attempts and analyze security events.




Server Manager
Server Manager ‣ Dashboard
Manage
Tools
View
Help
Dashboa
Local Ser
All Serve
AD DS
DNS
File and S
New Object - User
Create in: lab.local/Users
Password:
Confirm password:
User must change password at next logon
User cannot change password
Password never expires
Account is disabled
< Back
Next >
Cancel
Group Polic... Security Group... Members in this group c...
Guest User Built-in account for gue...
Key Admins Security Group... Members of this group ...
Protected Us... Security Group... Members of this group ...
RAS and IAS ... Security Group... Servers in this group can...
Read-only D... Security Group... Members of this group ...
Hide
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Services
Performance
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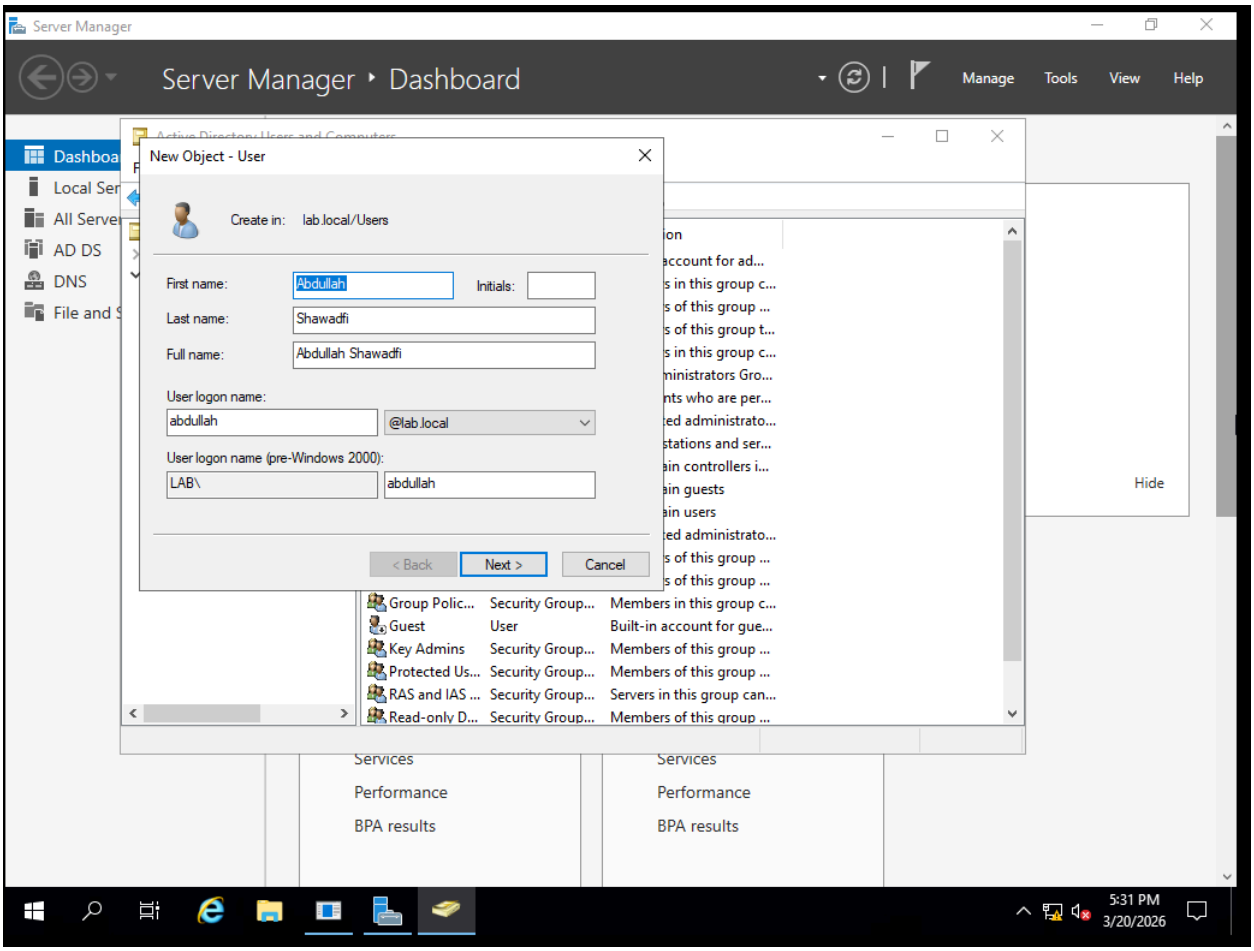


This image shows the Windows 11 client running commands such as whoami, hostname, and ping to verify system identity and network connectivity with the Active Directory server.

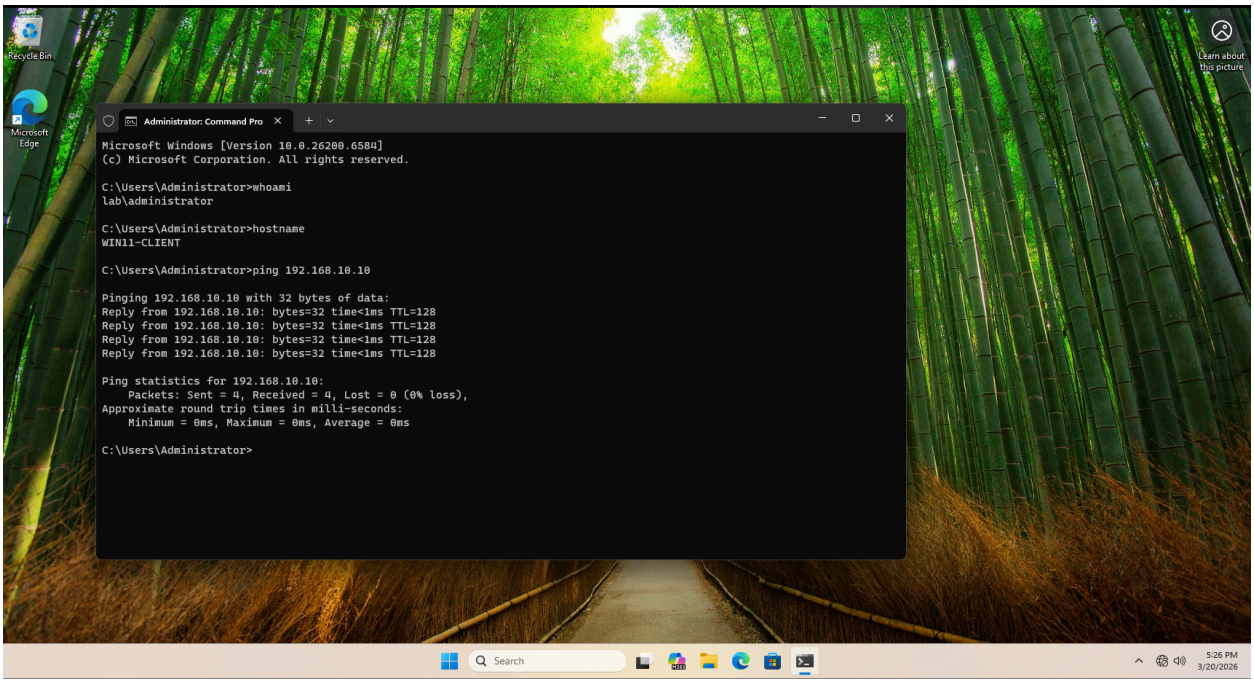


Figure 2: Failed login attempts (Event ID 4625)
This shows multiple authentication failures for the same account

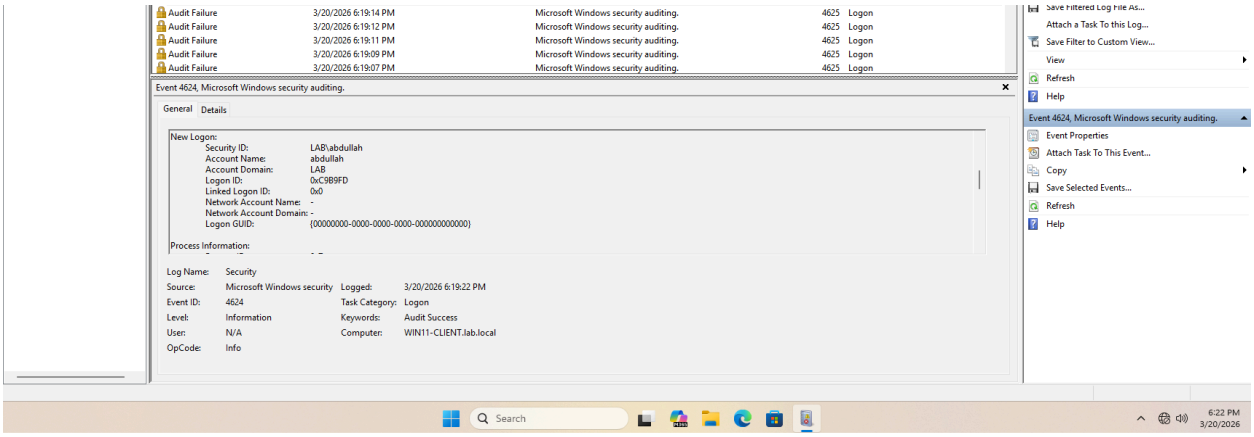
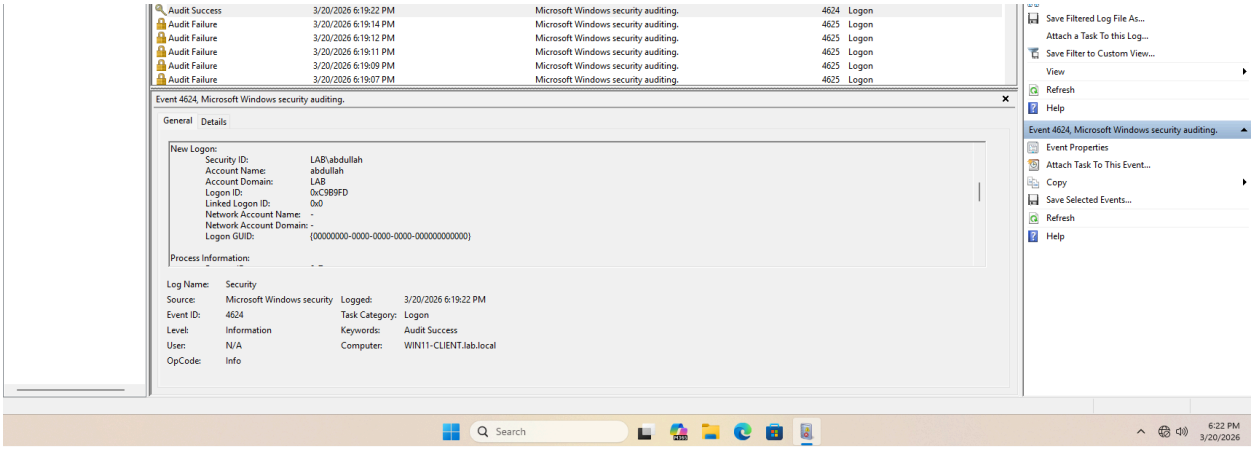


Figure 3: Successful login (Event ID 4624)
This confirms that the login was eventually successful after several failed

attempts.



🔹 Analysis

The analysis of Windows Security logs revealed multiple failed login attempts (Event ID 4625) targeting the user "Abdullah".

The source IP address associated with these attempts was 127.0.0.1, which indicates that the login attempts originated from the local machine itself (localhost), rather than an external system.

Each failed login event showed a Status code of 0xC000006A, which corresponds to an incorrect password attempt. This confirms that the authentication failures were due to invalid credentials.

Additionally, the Logon Type observed was Type 2 (Interactive), indicating that the login attempts were performed locally on the system, such as through the console or direct user interaction.

The high number of failed login attempts within a short period suggests repeated manual or automated password guessing attempts on the same machine.

Following these failed attempts, a successful login event (Event ID 4624) was recorded for the same user account.

This successful login also originated from 127.0.0.1 using Logon Type 2, confirming that access was eventually gained locally after multiple failed attempts.

This pattern of repeated failed logins followed by a successful login is consistent with a brute force attack scenario, even though it was performed locally within the system.

Such behavior may indicate either a user repeatedly attempting incorrect passwords or a local brute force simulation using scripts or tools.

Since the source IP is localhost (127.0.0.1), this activity is more likely part of a controlled lab simulation rather than a real external attack.

🔹 Detection Rule

An alert can be generated if:

- More than 5 failed login attempts (Event ID 4625)
- Followed by a successful login (Event ID 4624)
- Within a short time window (e.g., 2 minutes)

🔹 Dashboard

DashBoard.pdf

🔹 Conclusion

This project shows how brute force attacks can be identified using basic log analysis.

It highlights the importance of monitoring authentication logs to detect suspicious activity.